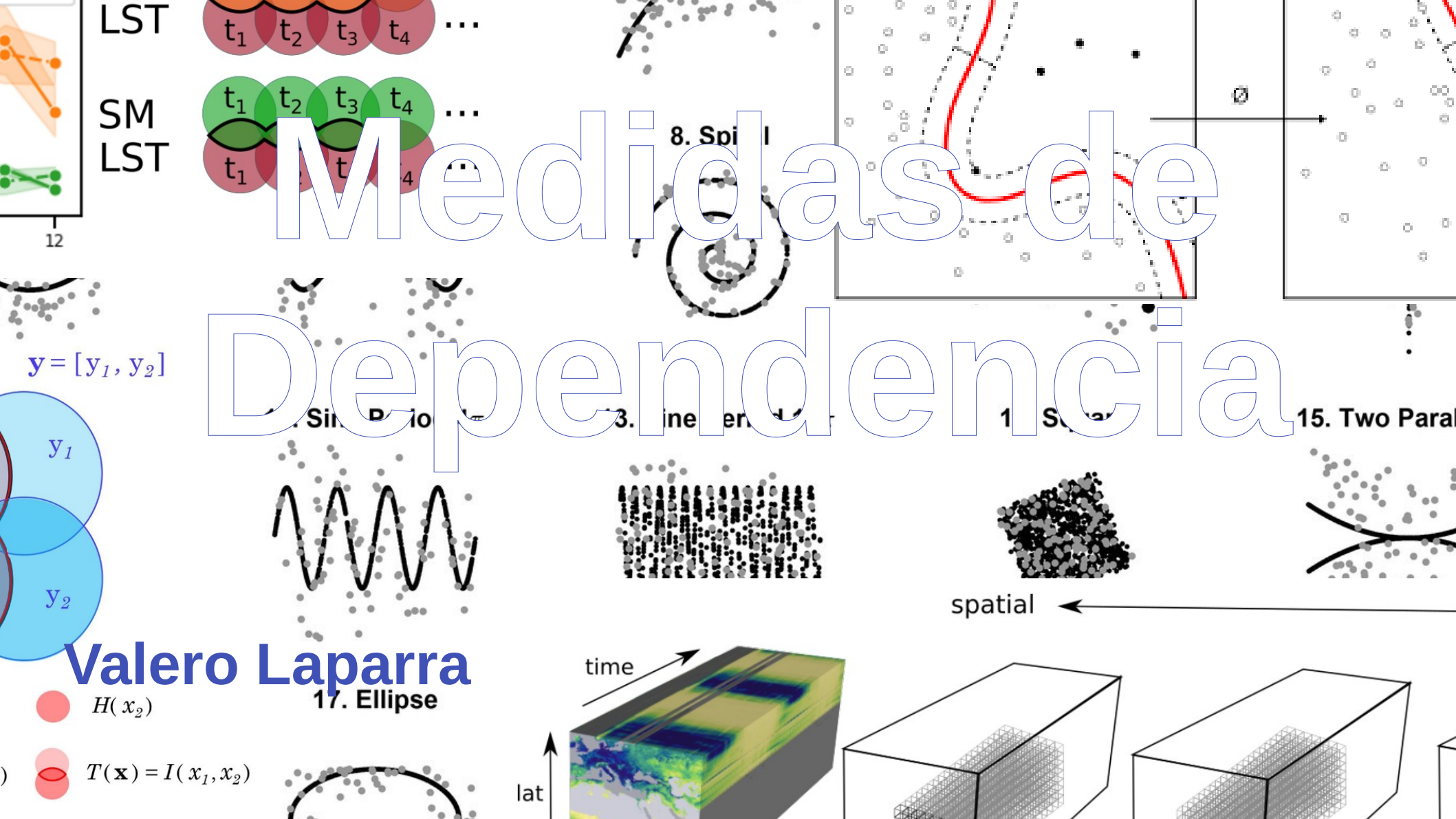
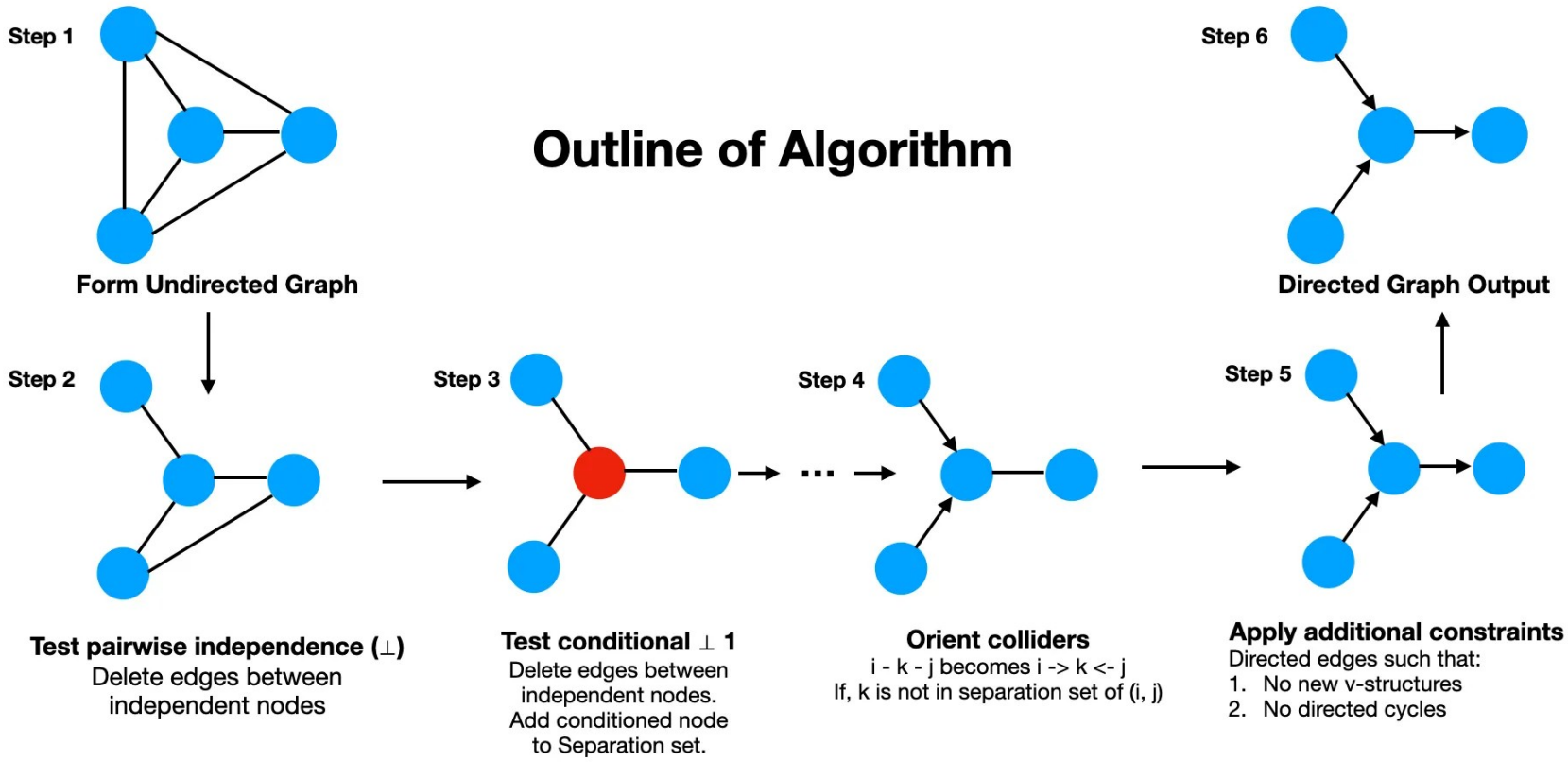
# Medidas de Dependencia

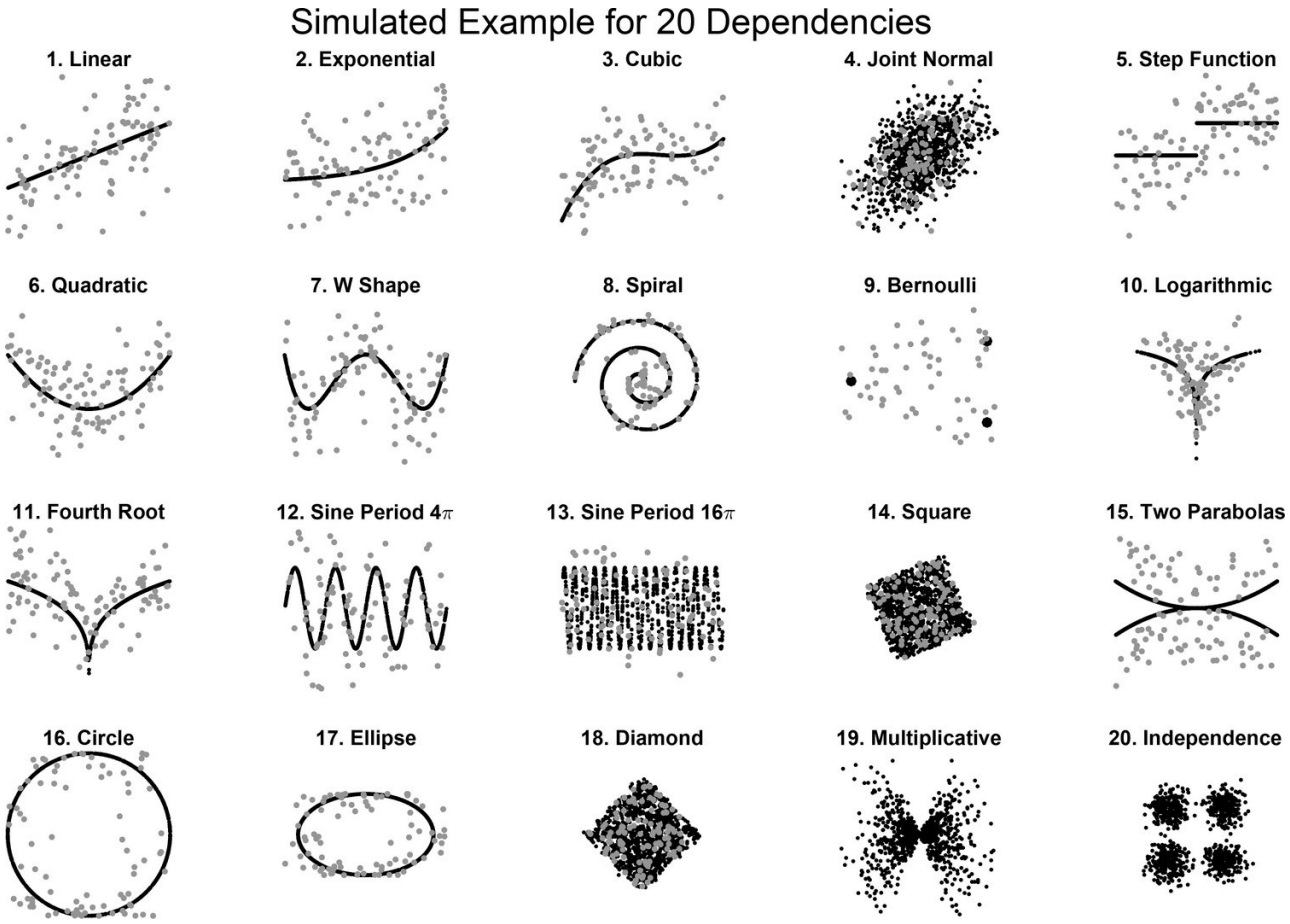
Valero Laparra



# Medidas de dependencia



# Medidas de dependencia



# Medidas de dependencia

## Correlación Pearson

$$\rho_{X,Y} = \operatorname{corr}(X,Y) = \frac{\operatorname{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\operatorname{E}[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$$

## Relación Lineal

# Medidas de dependencia

## Correlación Pearson

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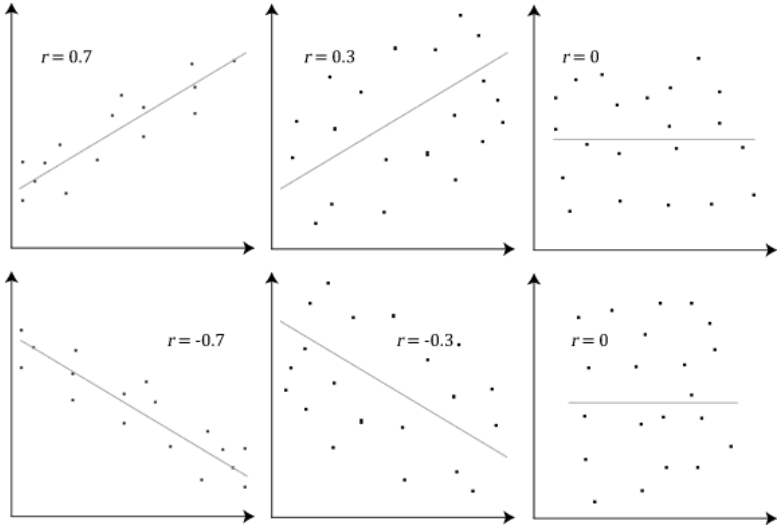


## Relación Lineal

# Medidas de dependencia

## Correlación Pearson

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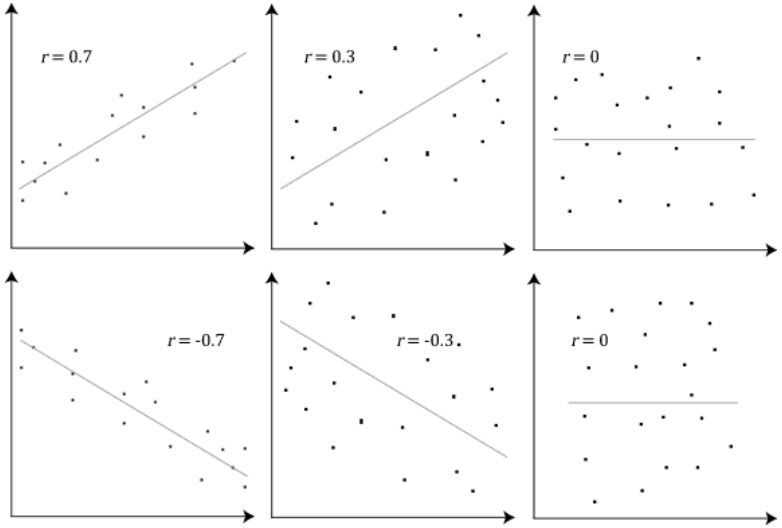



## Relación Lineal

http://guessthecorrelation.com/

# Medidas de dependencia

## Correlación Spearman

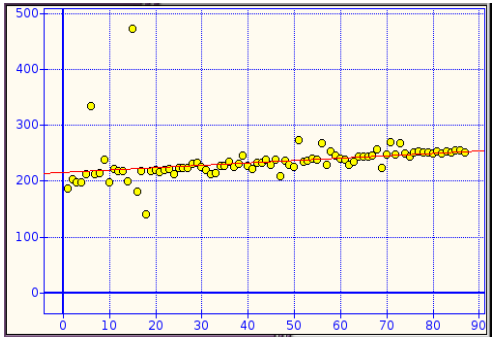
$$\rho = 1 - \frac{6 \sum D^2}{N(N^2 - 1)}$$

Relación orden

# Medidas de dependencia

## Correlación Spearman

$$\rho = 1 - \frac{6 \sum D^2}{N(N^2 - 1)}$$



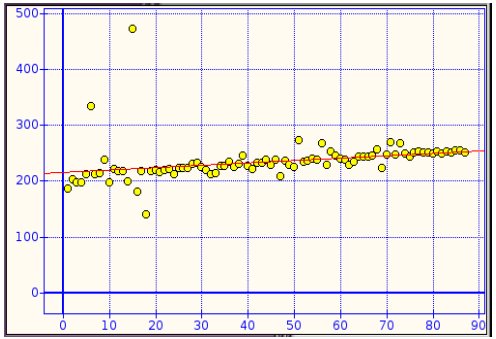

Pearson = 0.30 Spearman = 0.76

Relación orden

# Medidas de dependencia

## Correlación Spearman

$$\rho = 1 - \frac{6 \sum D^2}{N(N^2 - 1)}$$

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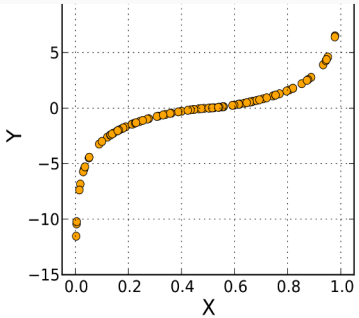


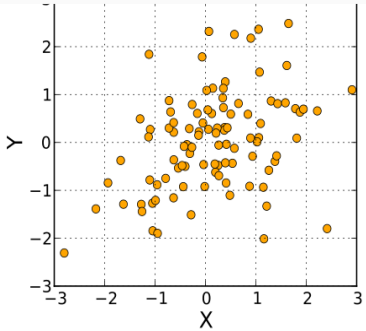


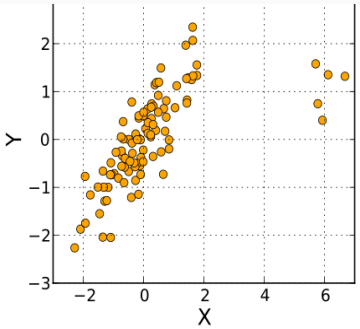

# Medidas de dependencia

## Correlación Spearman

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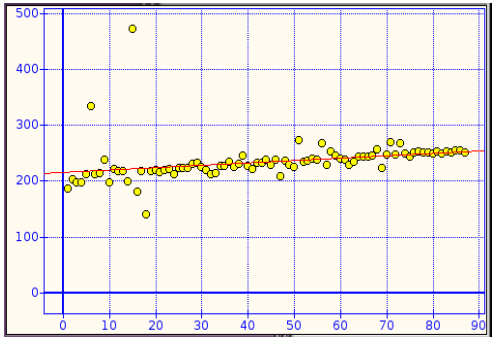


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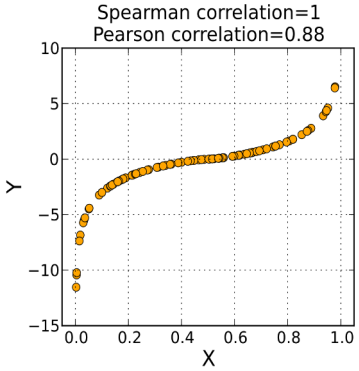
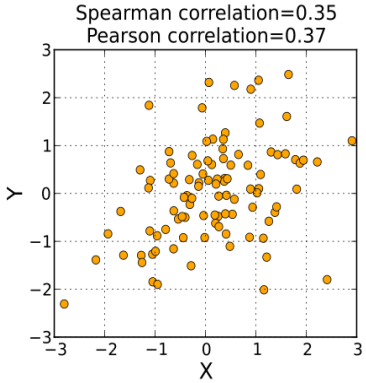
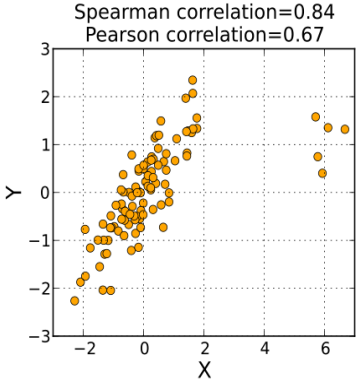


# Medidas de dependencia

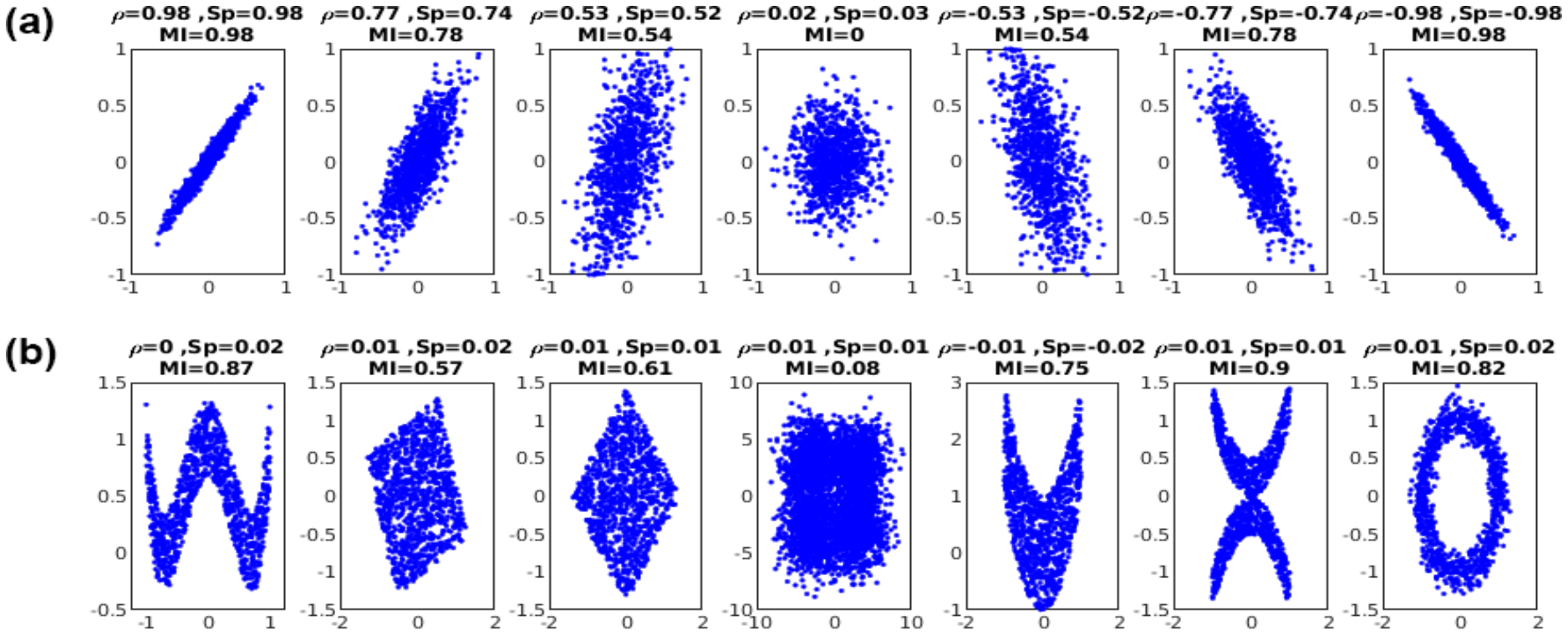
## Independecia

$$P(A \mid B) = P(A),$$

$$P(A \cap B) = P(A)P(B)$$

# Medidas de dependencia

## Independecia



# Medidas de dependencia

**Teoría de la información**

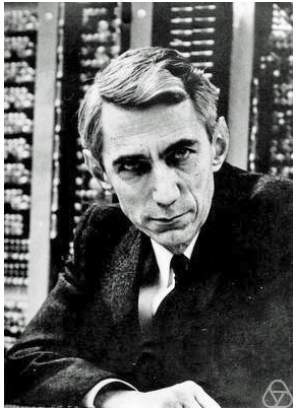
# Medidas de dependencia

**¿Qué es Información?**

# Medidas de dependencia

**Información**

Claude E. Shannon (1948) "A Mathematical Theory of Communication"



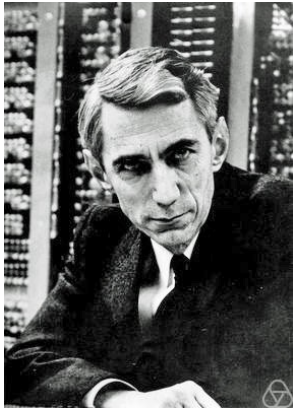
Información es proporcional a la inversa de la probabilidad

$$I(x) \propto \frac{1}{p}(x)$$

# Medidas de dependencia

**Información**

Claude E. Shannon (1948) "A Mathematical Theory of Communication"



Información es proporcional a la inversa de la probabilidad

$$I(x) = \log \frac{1}{p}(x)$$

# Medidas de dependencia

**Valor promedio de Información**

Entropía

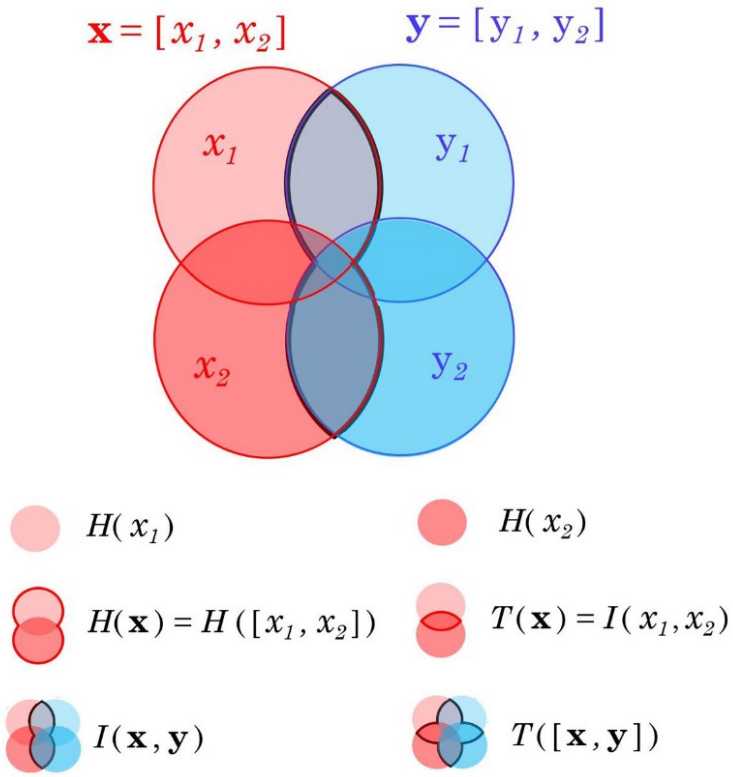
$$\mathrm{H}(X) = -\sum_{i=1}^{n} \mathrm{P}(x_i) \log \mathrm{P}(x_i)$$

Guardar datos (se mide en bits!)

# Medidas de dependencia

**Información mutua**

$$\mathrm{I}(X;Y) = \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} p_{(X,Y)}(x,y) \log \left( \frac{p_{(X,Y)}(x,y)}{p_X(x)\, p_Y(y)} \right),$$



# Medidas de dependencia

## ICC (Informatión Correlatión Coefficient)

Convierte la MI para poder ser interpretada como un cofficiente de correlación.

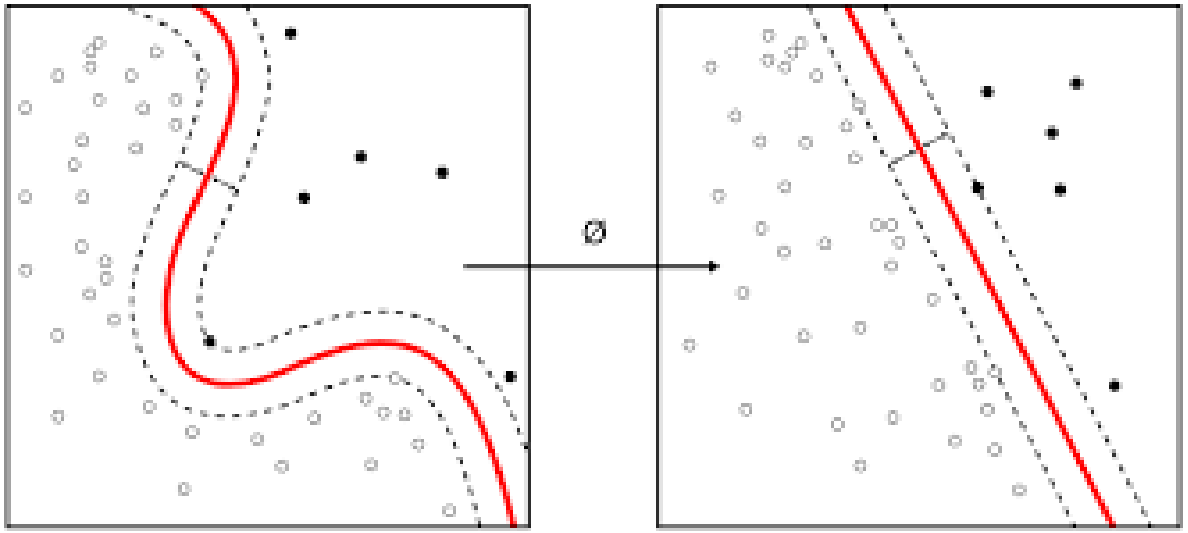
$$R(X,Y)=\sqrt{1-e^{-2I(X,Y)}}$$

**Information correlation coefficient**
https://encyclopediaofmath.org/wiki/Information_correlation_coefficient

**“An informational measure of correlation.”**
E.H. Linfoot
Information and Control, 1957

# Medidas de dependencia

**Medidas con Kernels**



# Medidas de dependencia

## Medidas con Kernels

https://jejjohnson.github.io/hsic_alignment/

**HSIC** (centered kernel, unnormalized)

$$\mathrm{HSIC} = \frac{1}{n(n-1)} \langle K_x H, K_y H \rangle_F$$

# Medidas de dependencia

## Medidas con Kernels

https://jejjohnson.github.io/hsic_alignment/

Kernel Alignment (**KA**) (uncentered kernel, normalized)

$$\mathrm{KA} = \frac{\langle K_x, K_y \rangle_F}{||K_x||_F ||K_y||_F}$$

# Medidas de dependencia

## Medidas con Kernels

https://jejjohnson.github.io/hsic_alignment/

Centered Kernel Alignment (**CKA**) (centered kernel, normalized)

$$\mathrm{cKA} = \frac{\langle K_x H, K_y H \rangle_F}{||K_x H||_F ||K_y H||_F}$$

# Medidas de dependencia

**Independencia**

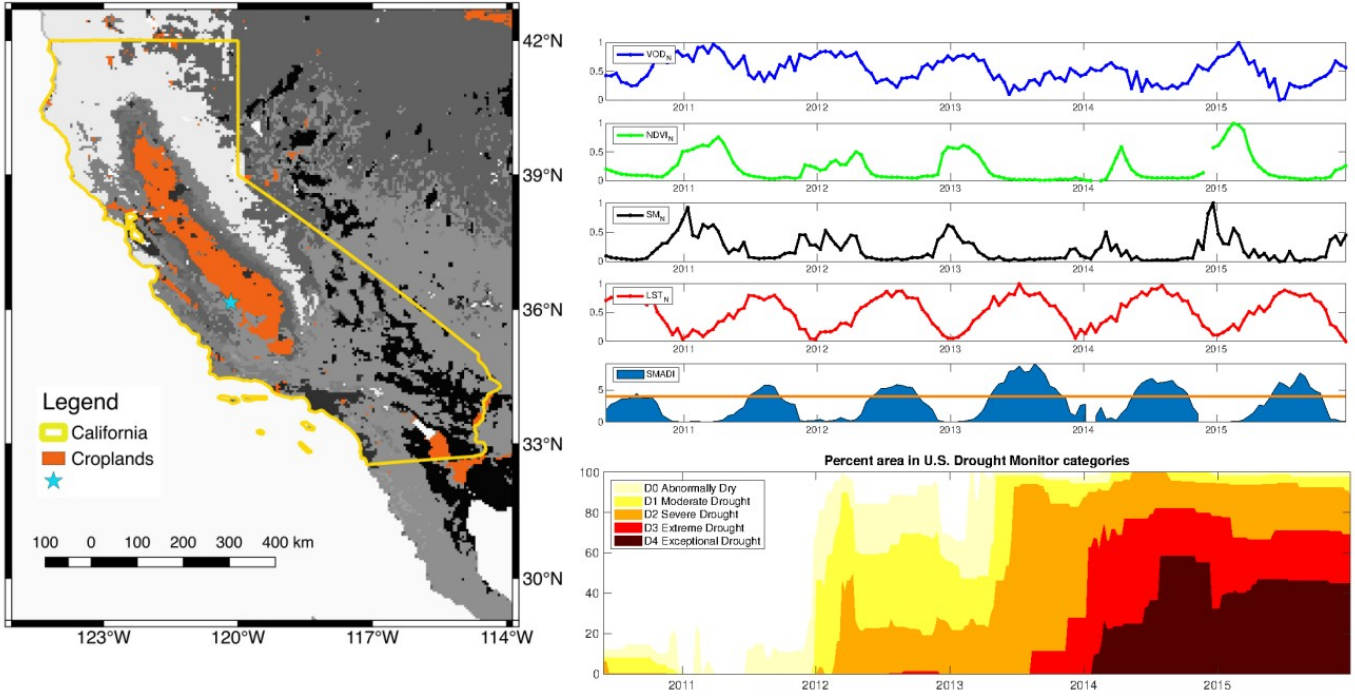
- Regresión: Correlaciones
  - Pearson
  - Spearman

- Probabilidad:
  - Información mutua
  - Kernels

# Medidas de dependencia

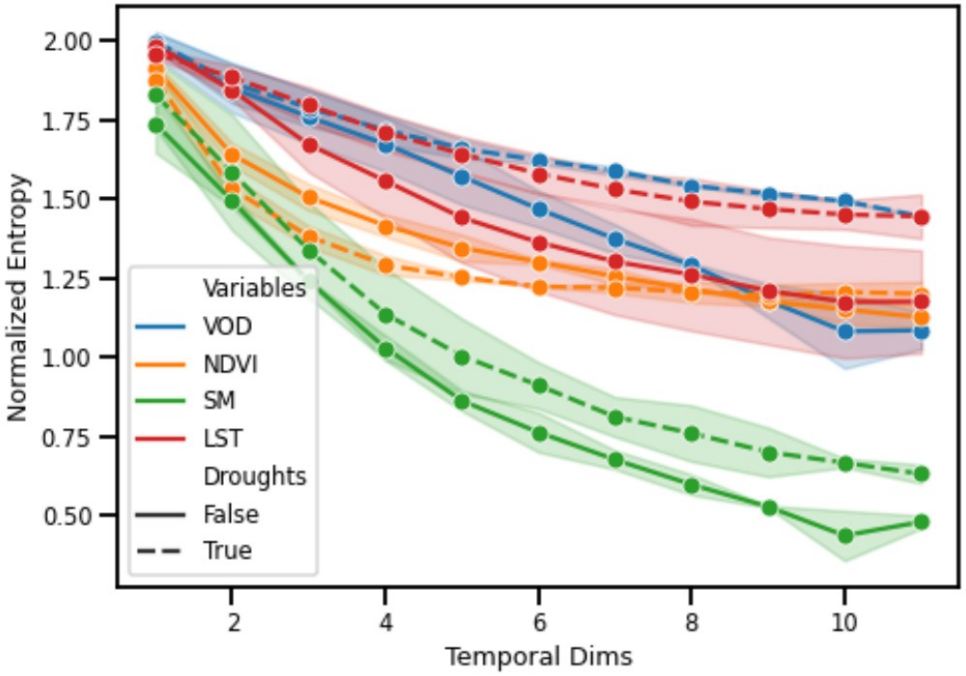
## Ejemplo en series temporales

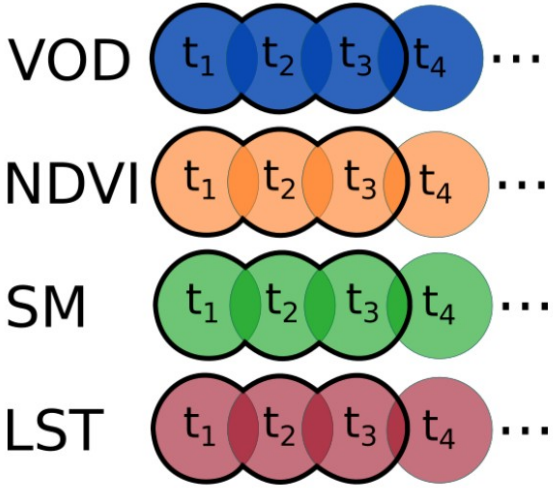
**Gaussianizing the Earth: Multidimensional Information Measures for Earth Data Analysis**
J Emmanuel Johnson, Valero Laparra, Maria Piles, Gustau Camps-Valls
IEEE Geoscience and Remote Sensing Magazine (2021)
Full Text : https://arxiv.org/pdf/2010.06476.pdf Python code : https://ipl-uv.github.io/rbig/



# Medidas de dependencia

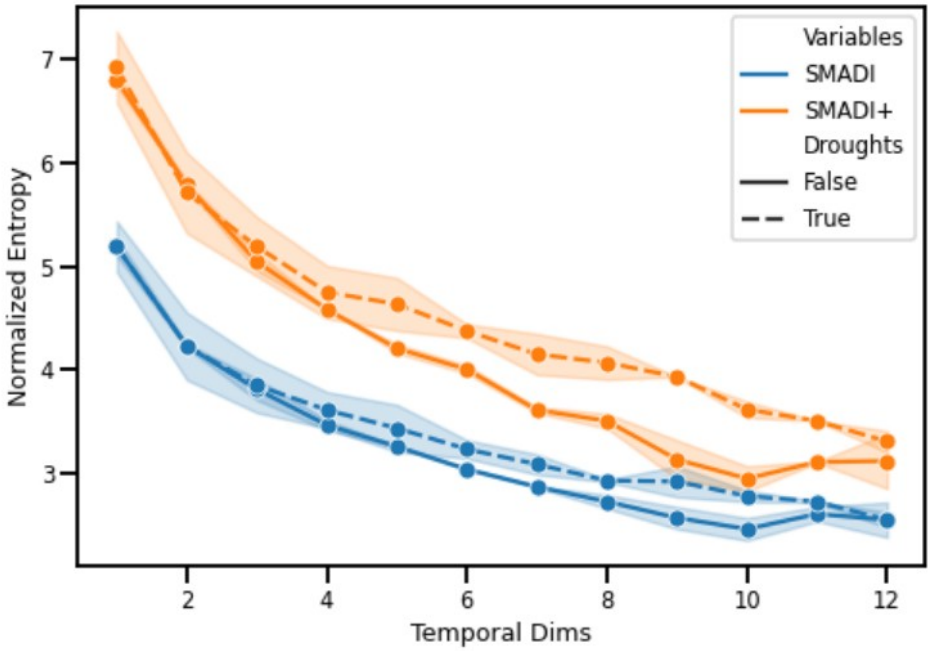
**Ejemplos en series temporales: ventana temporal**

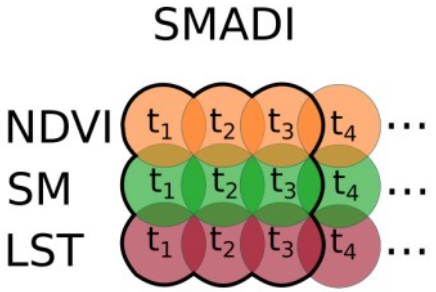


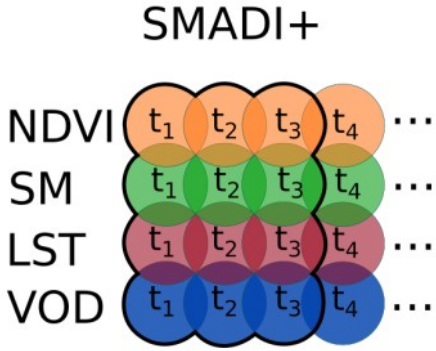



# Medidas de dependéncia

**Ejemplos en series temporales: comparación de modelos**

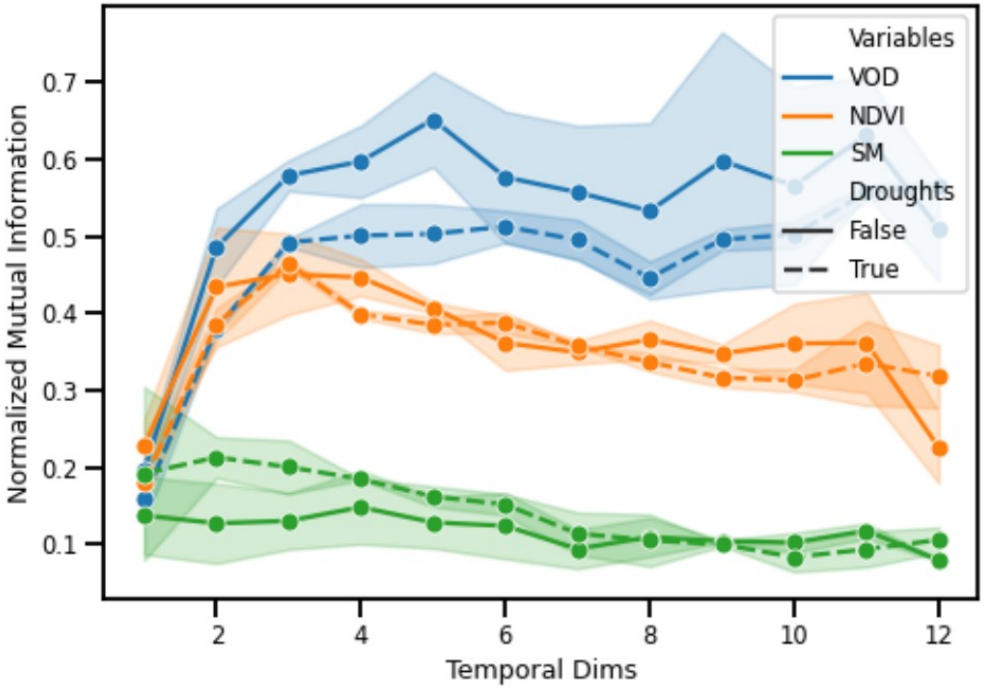


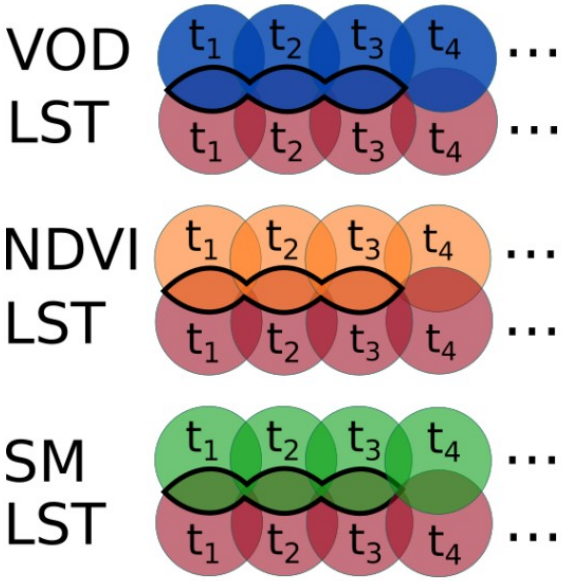



# Medidas de dependéncia

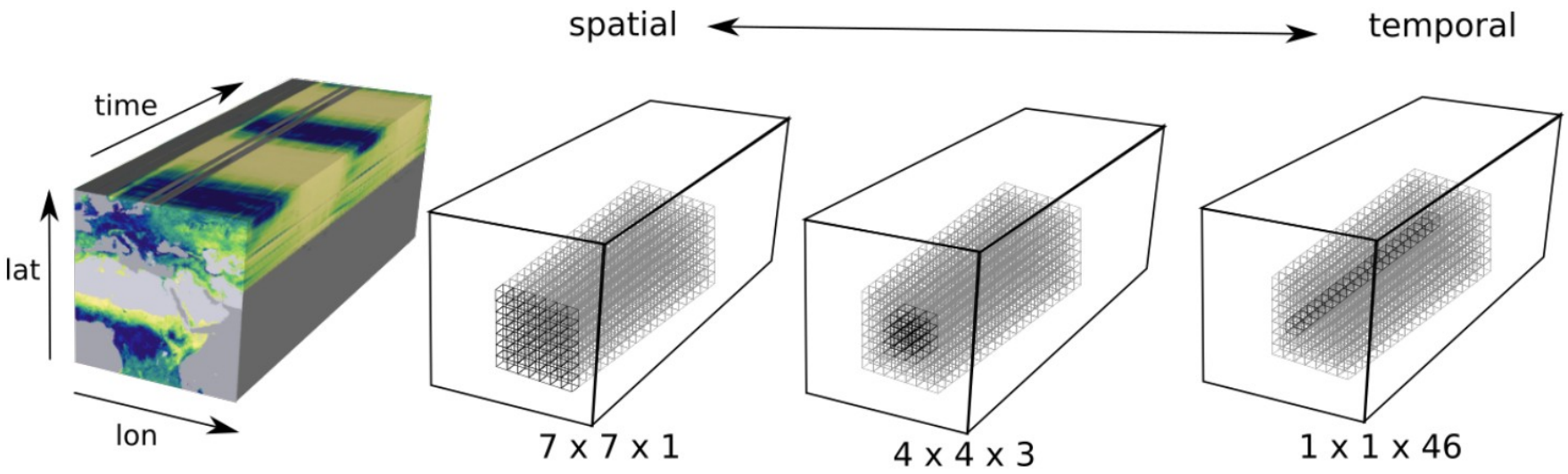
**Ejemplos en series temporales: redundancia entre variables**



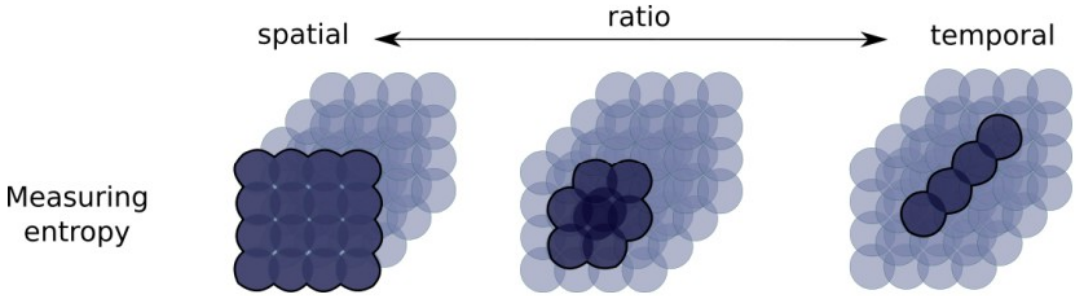
# Medidas de dependéncia

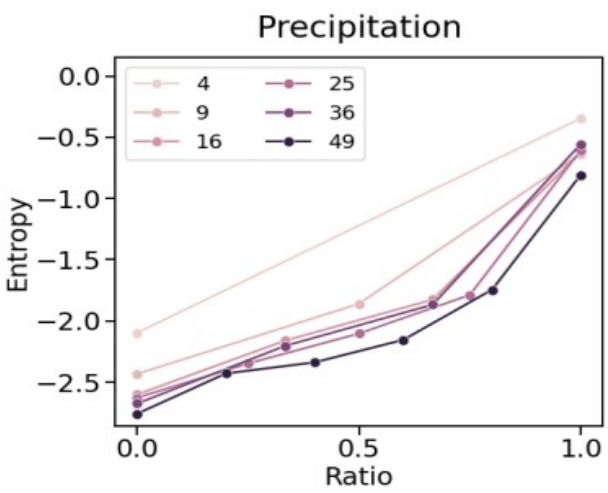
**Ejemplos espacio-temporal**

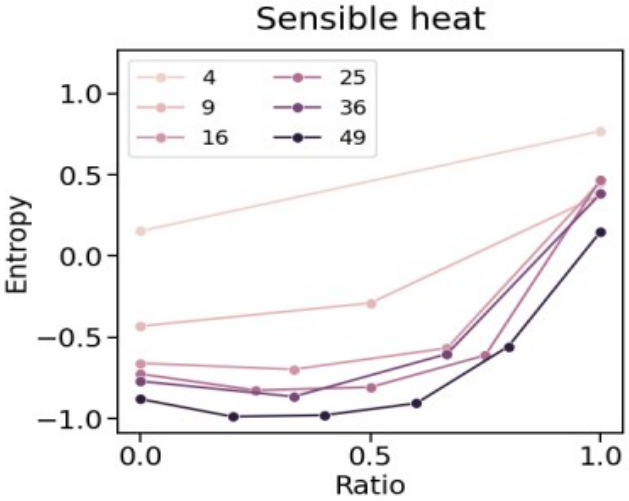


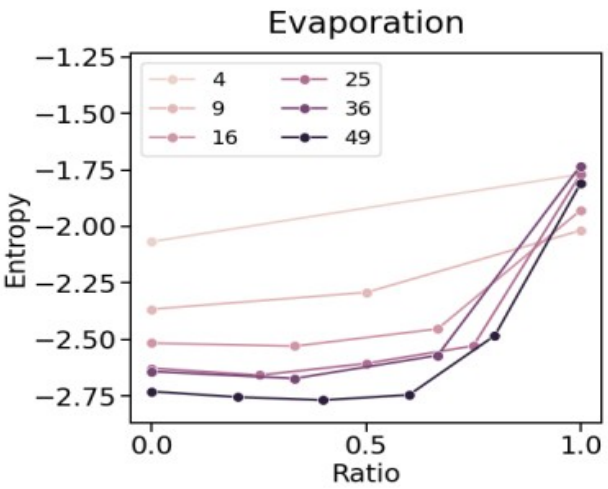
# Medidas de dependéncia

**Ejemplos espacio-temporal**



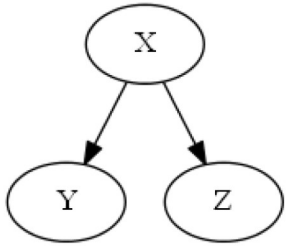
# Quitar conexiones

# Medidas de dependéncia

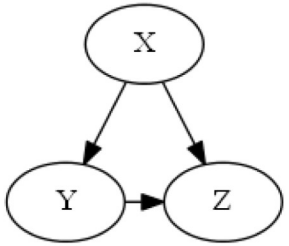
**Dependencia condicional: quitar conexiones en el grafo**

$$P(A|B) = \frac{P(A,B)}{P(B)}$$

$$(A \perp\!\!\!\perp B \mid C)$$




**(a)** $M_1 : Y \perp\!\!\!\perp Z|X$




**(b)** $M_2 : Y \not\perp\!\!\!\perp Z|X$

# Medidas de dependéncia

**Dependencia condicional: quitar conexiones en el grafo**

**- Test chi-square**

https://online.stat.psu.edu/stat504/book/export/html/730

**- Correlación parcial**

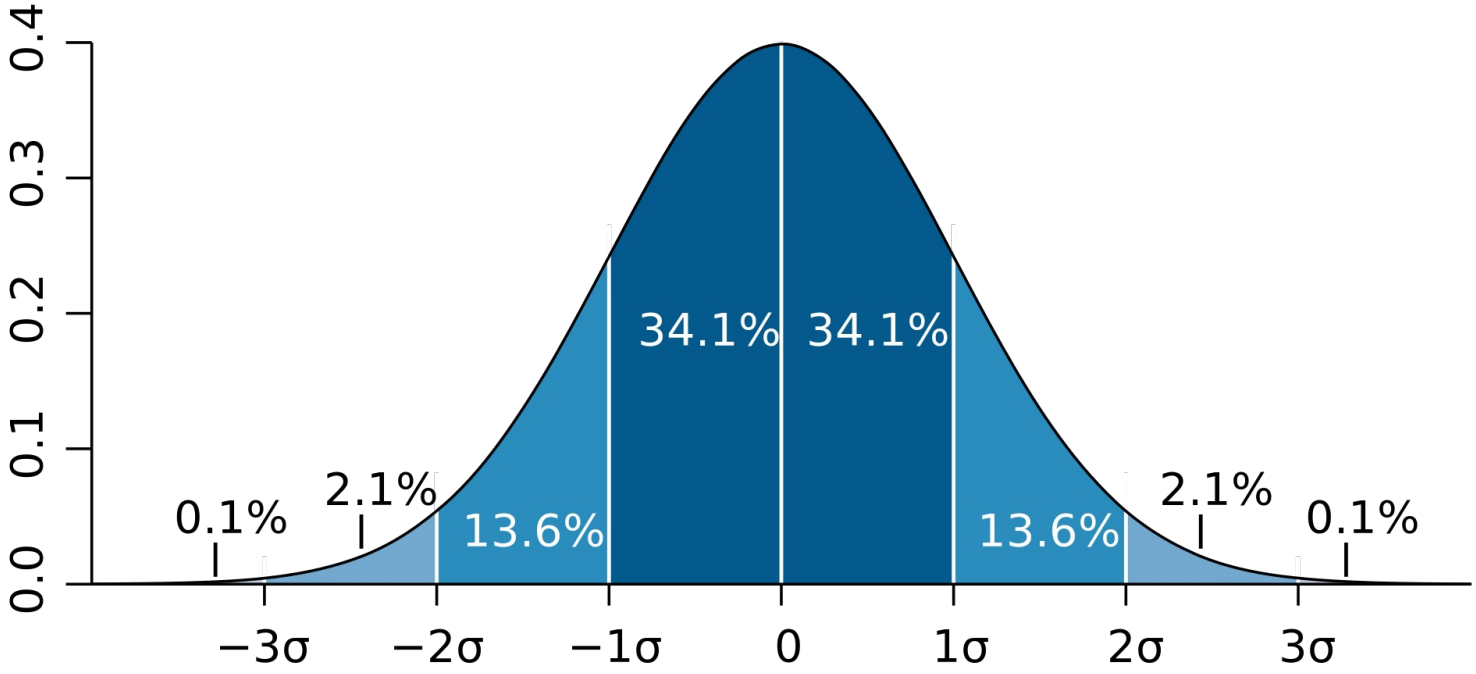
https://en.wikipedia.org/wiki/Partial_correlation

**- Información mutual condicional**

https://en.wikipedia.org/wiki/Conditional_mutual_information

# Medidas de dependéncia

**Intervalos de confianza**



# Medidas de dependéncia

**Tests de hipótesis**

Reducción al absurdo

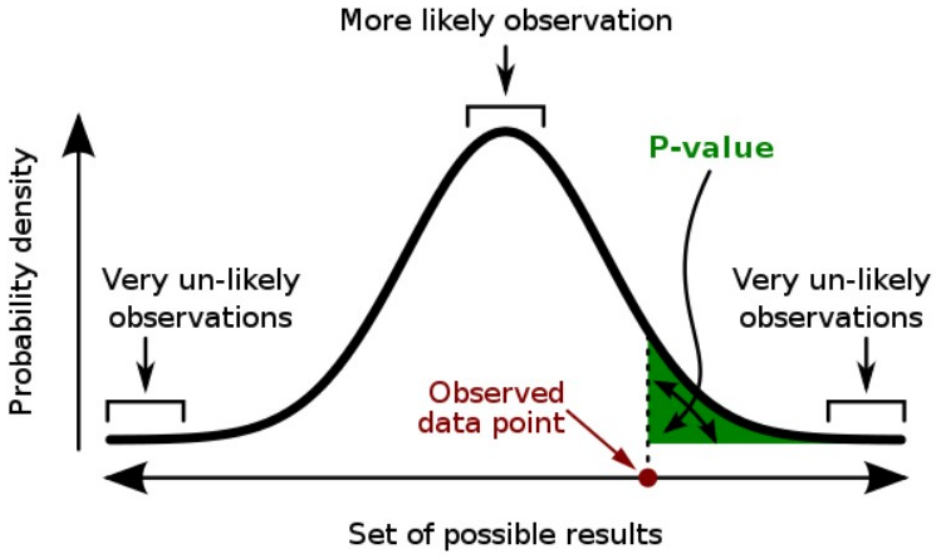
$H_0$ = hipótesis nula (no queremos que sea verdad)

$H_A$ = hipótesis alternativa (queremos que sea verdad)

# Medidas de dependéncia

## Tests de hipótesis

**P-valor**. Probabilidad de que un valor estadístico calculado sea posible dada una hipótesis nula cierta.



# Medidas de dependéncia

**Tests de hipótesis**

**P-valor**. Cuidado!

$$\boxed{P(\text{Observación}|\text{hipótesis})} \neq P(\text{hipótesis}|\text{observación})$$

# Poner flechas / Causalidad

# Medidas de dependéncia

The NEW ENGLAND JOURNAL of MEDICINE

OCCASIONAL NOTES FREE PREVIEW

## Chocolate Consumption, Cognitive Function, and Nobel Laureates

Franz H. Messerli, M.D.

Chocolate consumption could hypothetically improve cognitive function not only in individuals but in whole populations. Could there be a correlation between a country's level of chocolate consumption and its total number of Nobel laureates per capita?

October 18, 2012
N Engl J Med 2012; 367:1562-1564
DOI: 10.1056/NEJMon1211064

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"Chocolate consumption enhances cognitive function"

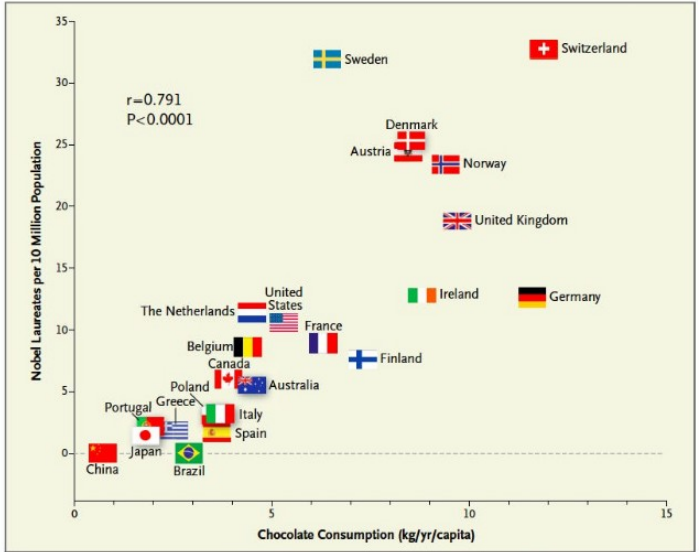


Figure 1. Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.

# Medidas de dependéncia

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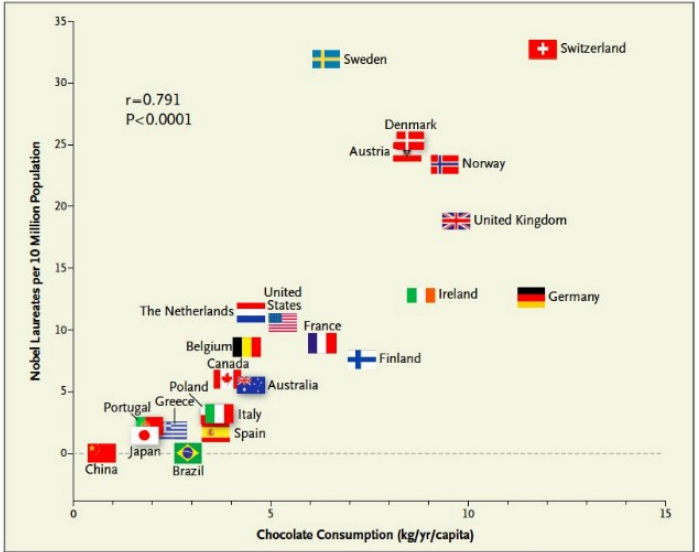
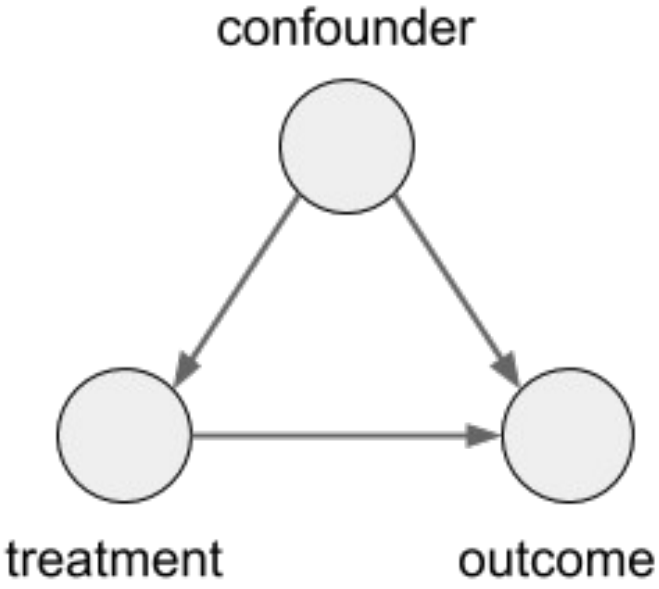



Figure 1. Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.

**Correlación (o incluso dependencia) no implica causalidad**

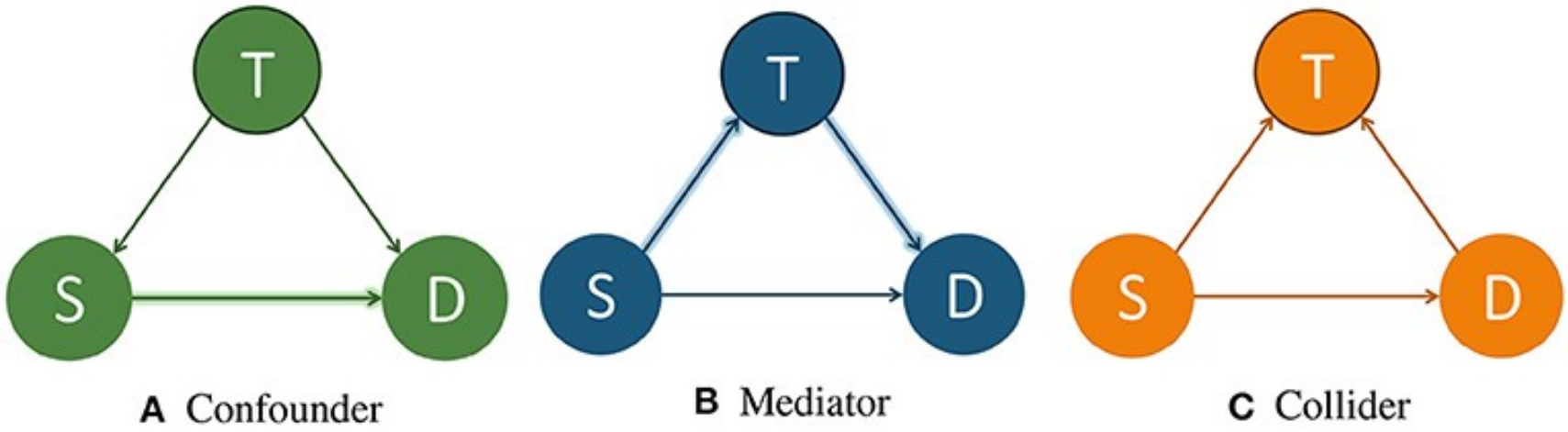
# Medidas de dependéncia




https://en.wikipedia.org/wiki/Exploratory_causal_analysis

# Medidas de dependéncia

**Dependencia condicional: orientar flechas**



# Medidas de dependéncia

**Medidas direccionales: Para poner flechas en el grafo**

**- Directed information**

https://en.wikipedia.org/wiki/Directed_information

**- Granger causality**

https://en.wikipedia.org/wiki/Granger_causality

**- Interaction information** (puede ser negativa):

https://en.wikipedia.org/wiki/Interaction_information

# Medidas de dependéncia

## Directional information

Universal Estimation of Directed Information
https://arxiv.org/pdf/1201.2334.pdf

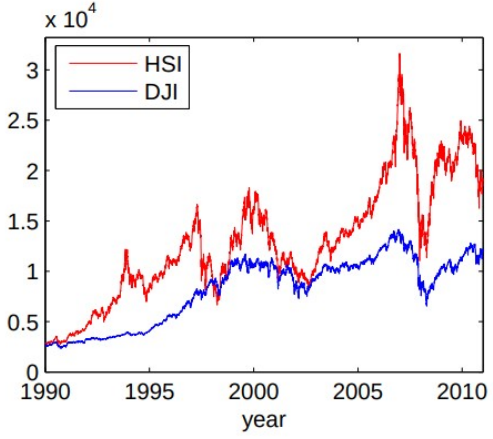



Fig. 9. The Hang Seng Index (HSI) and the Dow Jones Industrial Average (DJIA) between 1990 and 2011. The goal is to determine which index is causally influencing the other.

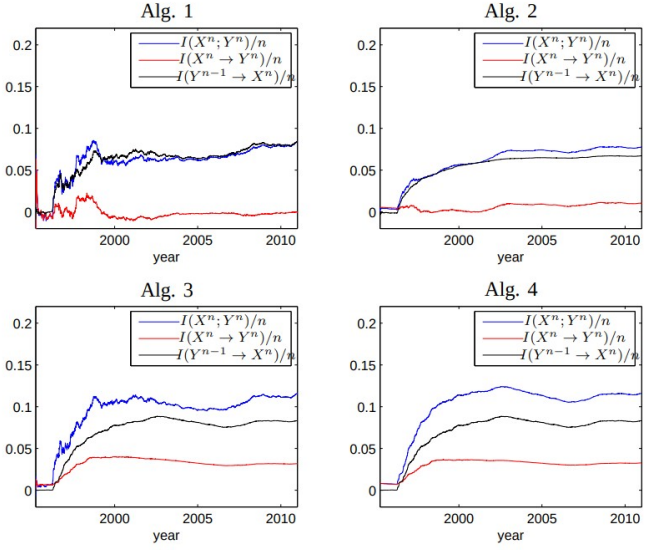
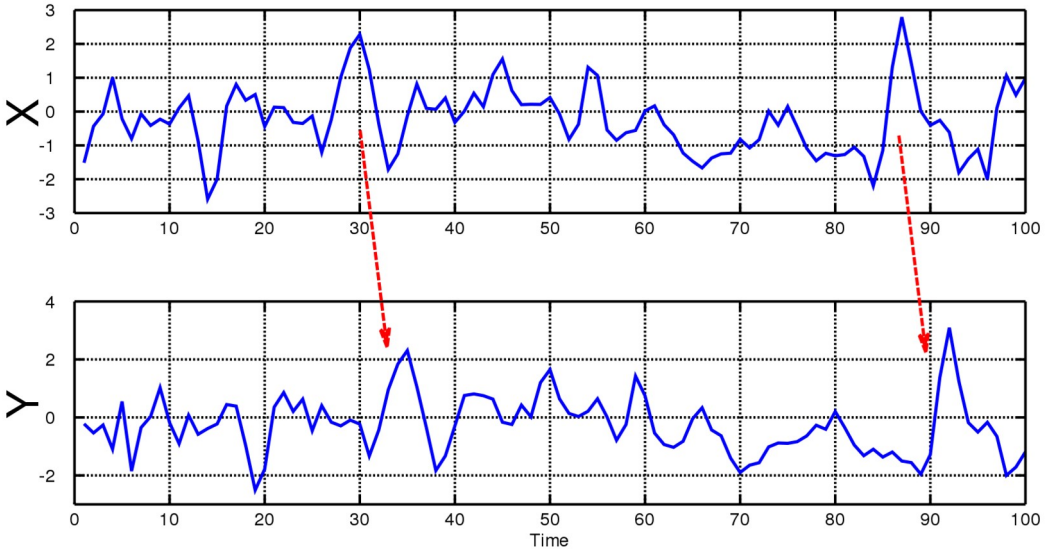



Fig. 10. Estimates of information-theoretic measures between HSI denoted by $\mathbf{X}$, and DJI denoted by $\mathbf{Y}$. It is clear that the reverse directed information is much higher than the directed information, hence it is DJI that causally influences HSI rather than the other way around.

# Medidas de dependéncia

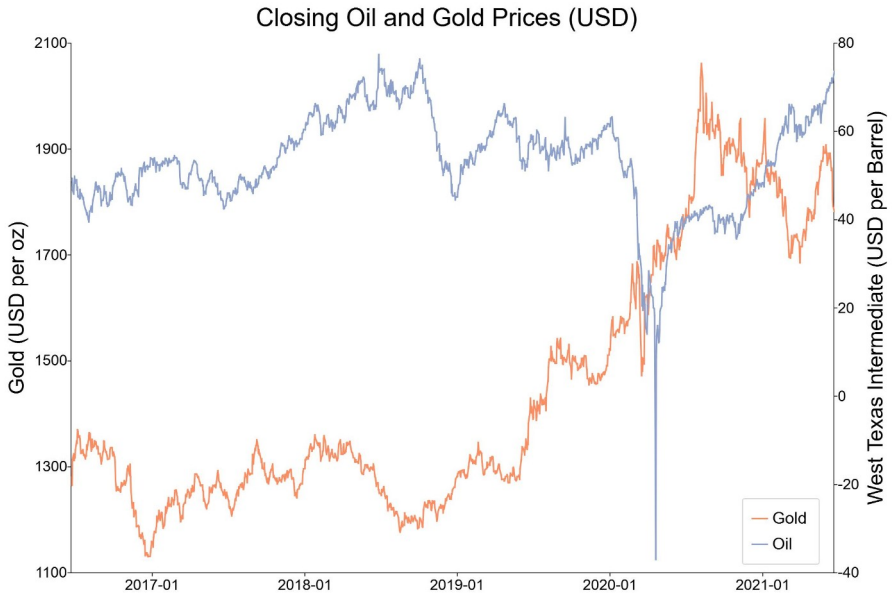
**De Granger**: a partir de regresión temporal



# Medidas de dependéncia

## Granger

https://www.aptech.com/blog/introduction-to-granger-causality/




```
   Standard Granger Causality Test
-------------------------------------------------------------
Direction                    Wald          Bootstrap p-val
price_oil => price_gold     6.860                 0.093
price_gold => price_oil     5.982                 0.109
```

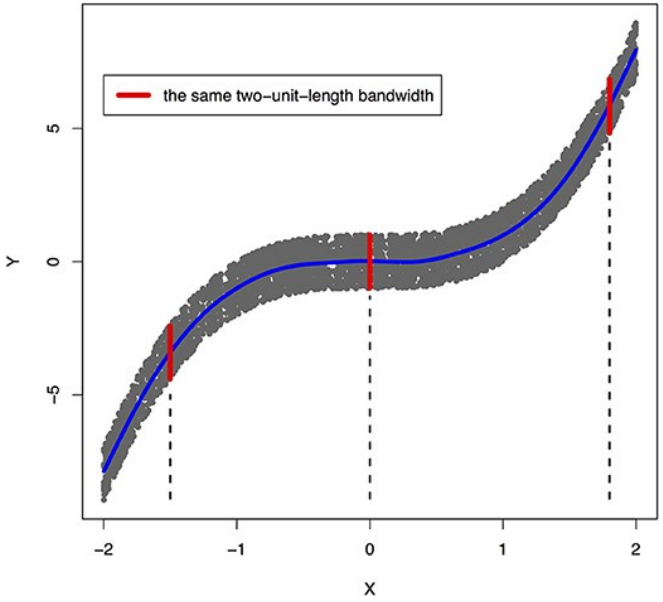
Rechazar la hipótesis nula de que los precios del petróleo no causan Granger a los precios del oro al nivel del 10%.

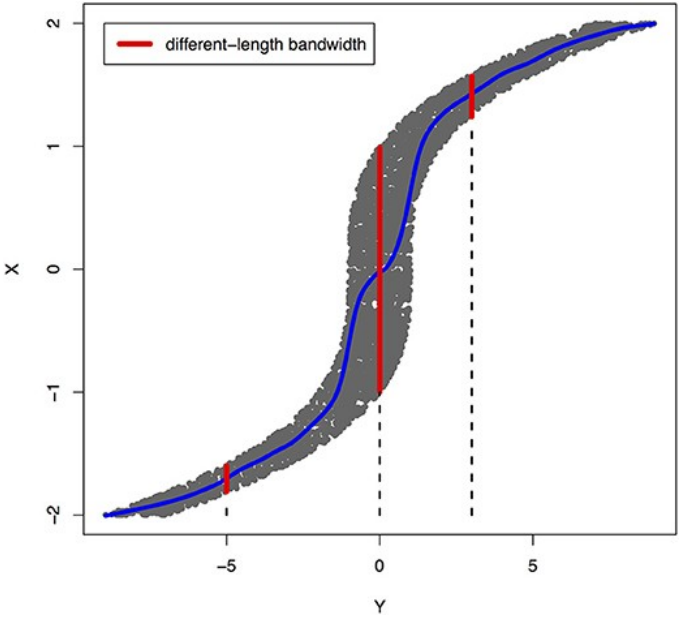
No se puede rechazar la hipótesis nula de que los precios del oro no son la causa de Granger de los precios del petróleo.

# Medidas de dependéncia

**Causality from residuals**

Asumir ruido aditivo: **y = f(x) + n** vs **x = f(y) + n**



# Medidas de dependéncia

## Causality from residuals

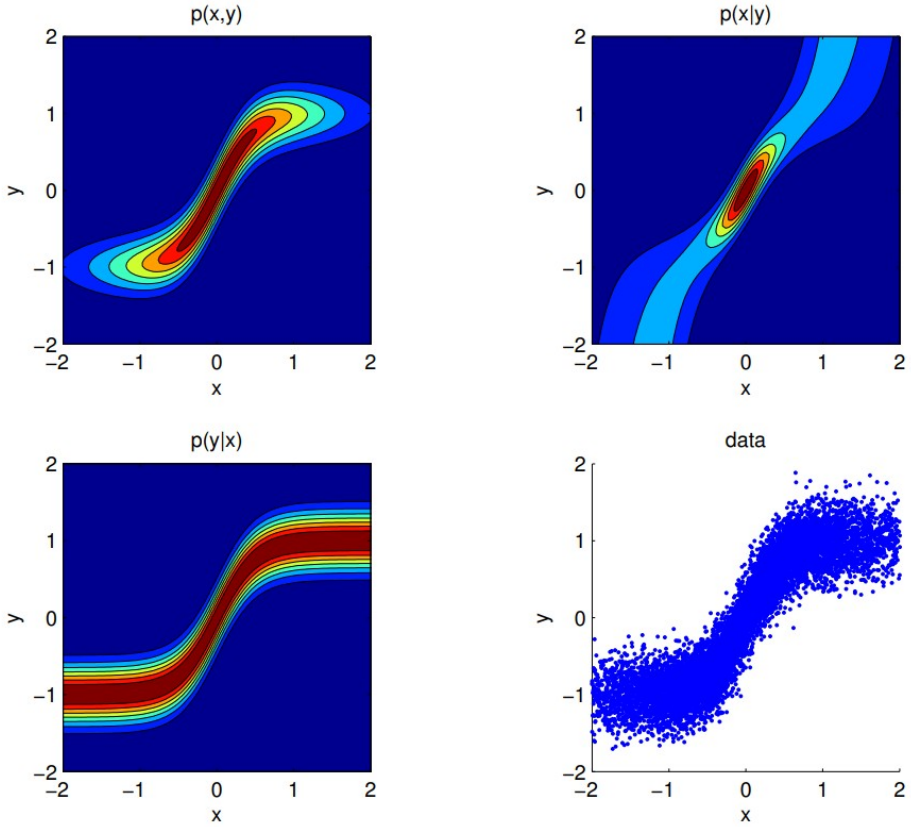
Asumir ruido aditivo:

**y = f(x) + n**

**x = f(y) + n**

”Distinguishing cause from effect using observational data: methods and benchmarks”
Mooij et al,

https://arxiv.org/pdf/1412.3773v1.pdf



# Medidas de dependéncia

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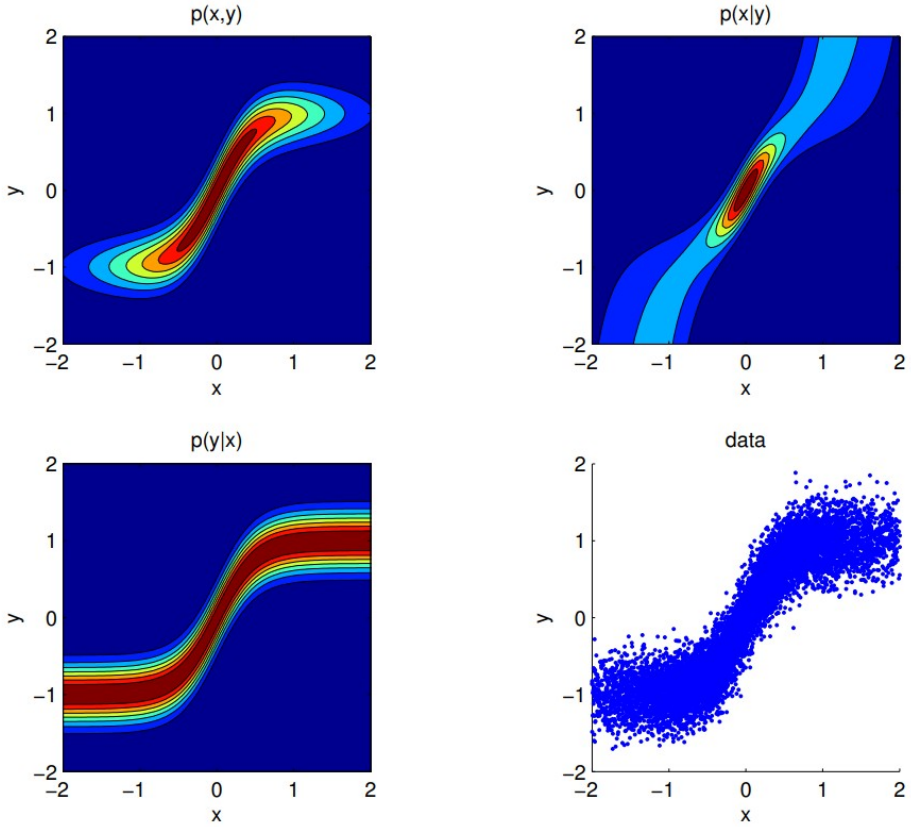
https://arxiv.org/pdf/1412.3773v1.pdf